

Bifurcations of Hamiltonian Periodic Orbits

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Well-known Classical Fact from 70s

There are 8 types of *generic* bifurcation of Hamiltonian periodic orbits.

Contents

- Dynamical Systems and Bifurcations
- Bifurcation and Symplectic Homology
- Ingredients : Classical works of Lyapunov and Meyer
- Classification of Generic Bifurcations

Main references : [Mey70], [AM78]

Dynamical Systems and Bifurcations

Toy Example : Critical Points of a Function

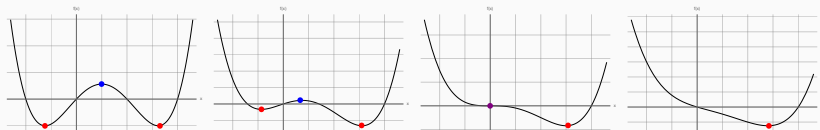


Figure 1: Graphs of $f_t(x) = x(x^2 - t)(x - 1)$ for varying t .

- $t < 0$, two minima (●) and one maximum. (●)
- $t = 0$, ● and ● meets at the inflection point. (●)
- $t > 0$, ● vanishes and one minimum survives. (●)

Idea. Similar thing happens for periodic orbits.

Example : Van der Pol Oscillator

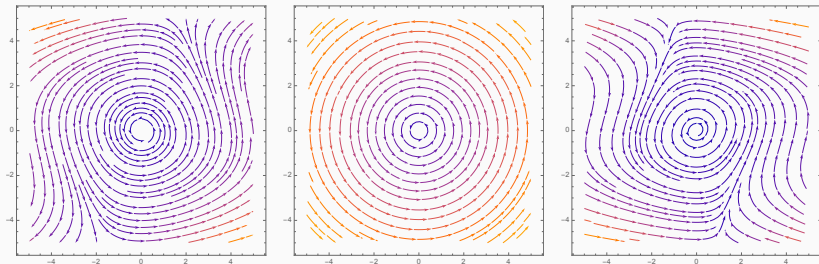


Figure 2: Phase portraits of Van der Pol oscillator for varying μ .

Van der Pol oscillator is given by

$$\dot{x} = \mu(1 - y^2)x - y, \quad \dot{y} = x.$$

This is an example of a phenomenon called Hopf bifurcation.

According to [Sma67], a **dynamical system** consists of following data.

- A space W (we use smooth manifold)
- A group action $G \rightarrow \text{End}(W)$ (we use $\text{Diff}(W)$)

If $G = \mathbb{Z} = \langle F \rangle$, we say the system is **discrete**.

Interesting objects

- **Fixed point** $x \in W$ such that $F(x) = x$.
- **Periodic point** $x \in W$ such that $F^N(x) = x$ for some N .
- **Invariant subspace** $S \subset W$ such that $F(S) \subset S$
- Non-wandering points, recurrent points, etc.

Continuous Dynamical System

If $G = \mathbb{R} = \{F_t\}$, we say the system is **continuous**.

A vector field $\frac{d}{dt}F_t|_{t=0} = X$ consists equivalent information with F_t , which is called the infinitesimal generator of F_t .

Example. The system determined by an ODE $\dot{x} = X(x)$.

Interesting objects

- **Critical points** $x \in W$ such that $F_t(x) = p$ for any t .
- **Periodic orbits** $x \in W$ such that $F_\tau(x) = x$ for some $\tau > 0$.
- **Invariant subsets** $S \subset W$ such that $F_t(S) \subset S$ for any t (or $t > 0$).

Hamiltonian Dynamical System

Let (W, ω) be a symplectic manifold, $H : W \rightarrow \mathbb{R}$.

H defines a **Hamiltonian vector field** X_H via relation

$$i_{X_H}\omega = \omega(X_H, -) = -dH.$$

We call (W, ω, H) a **Hamiltonian dynamical system**.

- X_H preserves ω , i.e. $\mathcal{L}_{X_H}\omega = 0$.
- X_H preserves H , i.e. X_H flows on $H^{-1}(c)$ for each c .

The first example is the geodesic flow on a Riemannian manifold (M, g) ,

$$H_g(q, p) = \frac{|p|_g^2}{2}$$

which is defined on T^*M .

Example : Classical Mechanics

Consider the mechanical system in $T^*\mathbb{R}^n$ with energy

$$H(q, p) = \frac{|p|^2}{2} + V(q).$$

Corresponding Hamiltonian vector field is

$$X_H = p \frac{\partial}{\partial q} - \nabla V \frac{\partial}{\partial p},$$

which can be translated into

$$\dot{q} = p, \quad \dot{p} = \ddot{q} = -\nabla V.$$

This is Newton's 2nd law, which governs the classical mechanics.

Structural Stability

Two vector fields X, Y on W are **topologically conjugate** if there exists a homeomorphism $F : W \rightarrow W$ which carries the oriented orbits of X to the ones of Y .

If X has a neighborhood $\mathcal{U} \subset \mathcal{X}(W)$ such that if $Y \in \mathcal{U}$ then X and Y are topologically conjugate, we call X is **structurally stable**.

Idea. A small perturbation of topologically stable vector field X cannot change the dynamical system essentially.

Bifurcation

Consider a family of vector fields X_t on W , parametrized by a set $0 \in I$.

A point $0 \in I$ is called

- a **robust point** if there exists a neighborhood U of 0 in I such that if $c \in U$, X_0 and X_c are topologically conjugate.
- a **bifurcation point** if not.

We will only consider 1-dimensional bifurcation, i.e. I is 1-manifold.

Idea. If we can find *generic* properties of vector fields, we can talk about *generic* bifurcations.

Bifurcation

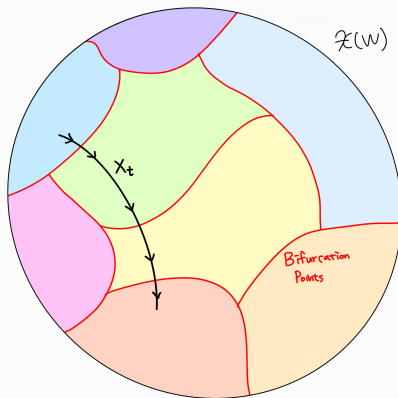


Figure 3: Very schematic picture of a bifurcation.

Bifurcation and Symplectic Homology

Theorem (Poincaré Section)

Let γ be a periodic orbit of X in W and $x \in \gamma$.

- 1. There exists a codimension 1 submanifold $S \subset W$ such that $x \in S$ and is transversal to X .*
- 2. Define a map $\Psi : S \rightarrow W$ by*

$$\Psi(q) = Fl_{\tau(y)}^X(y)$$

where $\tau(y) = \min\{t > 0 : Fl_t^X(y) \in S\}$, then this is well-defined and unique up to conjugation.

We call S **Poincaré section**, and Ψ **return map** or **Poincaré map**.

Poincaré Section

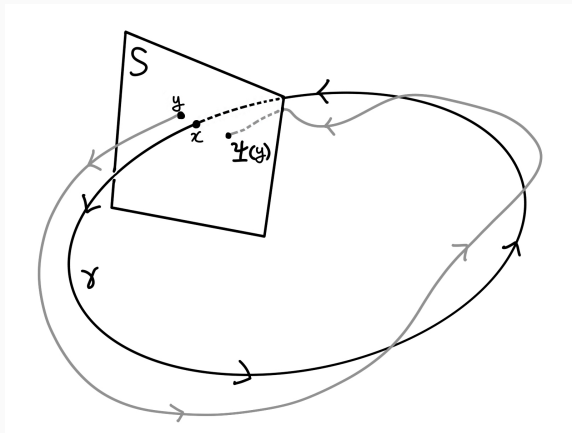


Figure 4: A Poincaré section and the return map

Hamiltonian Version of Poincaré Section

Proposition

Let $X = X_H$ be a Hamiltonian vector field and γ be a periodic orbit with regular energy c_0 . There exists a Poincaré section Y such that

- 1. $Y \subset W$ is a contact submanifold.*
- 2. $S_c = Y \cap H^{-1}(c)$ for c close to c_0 is a symplectic submanifold.*
- 3. The return map on S_c is a symplectomorphism.*

Idea. a periodic orbit of X_H on $W \Leftrightarrow$ a periodic point of Ψ on S_c .

Conley-Zehnder Indices

Let γ be a periodic Hamiltonian orbit of period τ , S be a Poincaré section, $\gamma \cap S = \{x\}$ and Ψ be the return map.

γ is **non-degenerate** if $\det(d\Psi|_x - \text{Id}) \neq 0$, i.e. the linearized return map doesn't have 1 as an eigenvalue.

The **Conley-Zehnder index** is an integer assigned to γ , defined by

$$\gamma_{CZ}(\gamma) = \frac{1}{2}\text{Sign}Q_0 + \sum \text{Sign}Q_t + \frac{1}{2}\text{Sign}Q_\tau \in \mathbb{Z}$$

where Q_t is a quadratic form $Q_t(v, v) = \omega(v, \dot{\Psi}(t)v)$ on $\ker(\Psi(t) - \text{Id})$, defined for t such that $\det(\Psi(t) - \text{Id}) = 0$, and Sign is the signature of a quadratic form.

(Very Brief) Symplectic Homology

$H : W \rightarrow \mathbb{R}$ be a Hamiltonian, c be a regular value.

Assume : $H^{-1}(c)$ is a contact type hypersurface, consists only of non-degenerate periodic orbits.

$\Rightarrow$ We can define **symplectic homology** $SH^{S^1,+}(W_c)$,

- generated by periodic Hamiltonian orbits on $H^{-1}(c)$,
- graded by Conley-Zehnder indices,
- Differentials are given by Floer cylinder.

Relation between Symplectic Homology and Bifurcation

If $\text{CritVal}(H) \cap [c_1, c_2] = \emptyset$, then $SH^{S^1,+}(W_{c_1}) \simeq SH^{S^1,+}(W_{c_2})$.

$\Rightarrow$ CZ-indices of *all* periodic orbits are topologically controlled.

Assume that we're away from the critical energies.

- Assume that there exists a family of periodic orbits γ_c parametrized by the energy c . (an *orbit cylinder*)
- If γ_c is non-degenerate for $c_1 \leq c \leq c_2$, then $\mu_{CZ}(\gamma_{c_1}) = \mu_{CZ}(\gamma_{c_2})$.
- Once γ_c degenerates, $\mu_{CZ}(\gamma_c)$ might change, or the family might just vanish away.
- Since $SH^{S^1,+}(W_c)$ must be preserved, other orbits must take the charge of the changed index by being born, vanishing, or changing its own index.

Schematic Diagram of Bifurcation and Index Change

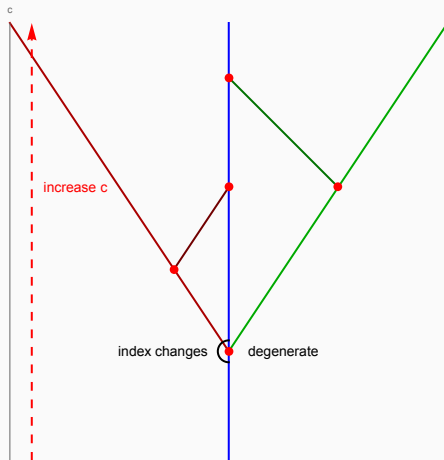


Figure 5: An example of bifurcation diagram. Each colored line indicates a family of periodic orbits, and $\bullet$ indicates the bifurcation and index-changing point.

Ingredients : Classical Works

Characteristic Exponents of Critical Point

Let x be a critical point of X_H on W^{2n} .

The **characteristic exponents** of x are the eigenvalues of $dX_H(p)$.

x is called **$\mathcal{H}$ -elementary** if

1. $0 \notin CE(x)$.
2. If $\alpha \in CE(x) \cap i\mathbb{R}$, then λ has multiplicity 1.
3. If $\alpha, \beta \in CE(x) \cap i\mathbb{R}_{>0}$, then they are $\mathbb{Z}$ -linearly independent.

Theorem ([Buc70])

For generic Hamiltonian H , every critical point of X_H is $\mathcal{H}$ -elementary.

Lyapunov's Theorem

Theorem (Lyapunov)

Let x be an $\mathcal{H}$ -elementary critical point of X_H . If $i\beta \in CE(x)$ for $\beta \in \mathbb{R}$, there exists a 2-dimensional submanifold C_β such that

- 1. $x \in C_\beta$.*
- 2. C_β is X_H -invariant submanifold.*
- 3. $T_x C_\beta$ is an eigenspace corresponds to $\pm i\beta$.*
- 4. C_β is a union of periodic orbits γ_r which converges to x as $r \rightarrow 0$.*
- 5. The period of γ_r converges to $2\pi/\beta$ as $r \rightarrow 0$.*

Lyapunov Orbits

Example. Lyapunov orbits of the restricted three-body problem, emerges from the Lagrange points.

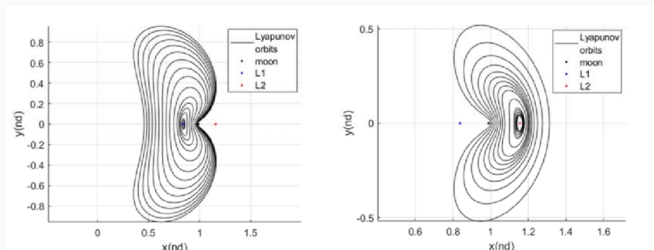


Figure 6: [ZWZZ20] Lyapunov orbits of RTBP, associated to L_1 and L_2 .

Reduction to Dimension 2

Interest : Bifurcation of a Hamiltonian orbit γ on symplectic 4-manifold.

Instead, we consider a periodic point of symplectomorphism on D^2 , which is a Poincaré section of γ and the return map.

By identifying D^2 's for various energy levels, we have following setting.

- M : 2-dimensional symplectic manifold.
- φ_t : 1-parameter family of symplectomorphisms parametrized by S .

We say (x, s) is fixed point if x is a fixed point of φ_s .

Characteristic Multiplier of Periodic Orbits

The **characteristic multiplier** of (x, s) is the eigenvalue λ_x of

$$d\varphi_s : T_x M \rightarrow T_x M,$$

which is chosen to be $1, -1, \operatorname{Im}(\lambda) > 0$ or $|\lambda| > 1$.

The characteristic multiplier λ_γ of corresponding periodic orbit γ is defined in the same way.

- $d\varphi_s$ is symplectic, so the eigenvalues are $\lambda^{\pm 1}$.
- If $|\lambda_x| > 1$, so $\lambda_x \in \mathbb{R}$, call x or γ **hyperbolic**.
- If $|\lambda_x| = 1$ and $\lambda \neq \pm 1$, call x or γ **elliptic**.

Note. The characteristic exponent and multiplier are different, and related by exponential.

Elliptic and Hyperbolic Fixed Points

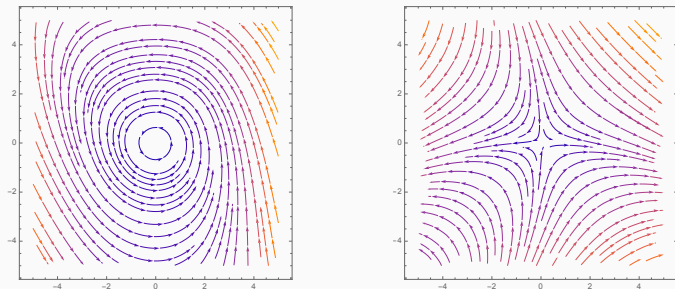


Figure 7: Elliptic and hyperbolic fixed points

In the phase portrait, an elliptic fixed point or orbit can be detected by nearby orbits. (KAM torus)

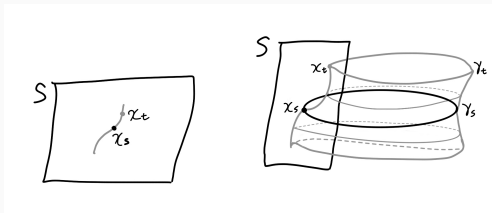
But the hyperbolic one cannot be detected from the nearby orbits.

In this sense, we say hyperbolic orbits are *quantatively invisible*.

Orbit Cylinder

Lemma (Orbit Cylinder)

If $\lambda_x \neq 1$, there exists neighborhoods $x_s \in U \subset M$ and $x \in V \subset S^1$ and a smooth map $P : V \rightarrow U$ such that $P(s) = x$ and $\{(P(t), t) : t \in V\}$ consists of fixed points.



In dimension 4, this can be interpreted as a family of orbits, and the *height* parameter t can be chosen as the energy.

Note. This theorem holds for arbitrary dimension.

Generating Function

Locally, we take a fixed point $x = (0, 0)$ and denote φ_s as

$$Q = Q(q, p), \quad P = P(q, p).$$

[Poi87] observed that

$$\Omega = (P - p)d(Q + q) - (Q - q)d(P + p)$$

is closed, so there exists a function G such that $dG = \Omega$.

We call G a **generating function**.

Generating Function

Assume that $\lambda \neq -1$, then under new coordinates

$$\xi = Q + q, \quad \eta = P + p,$$

we have

$$\frac{\partial G}{\partial \xi} = P - p, \quad \frac{\partial G}{\partial \eta} = q - Q,$$

and the fixed point of (Q, P) corresponds to the critical points of G .

- Hyperbolic fixed point of φ_s corresponds to a saddle point of G .
- Elliptic fixed point of φ_s corresponds to maximum or minimum of G .

(See Appendix 1 for the reason.)

Meyer's Theorem - Extremal Fixed Point

We call (x, s) is **extremal fixed point** if, for generating function $G = G(q, p, t)$, at $(0, 0, 0)$,

$$G = G_q = G_p = G_{pp} = G_{qp} = 0, \\ G_{qq} \neq 0, \quad G_{pt} \neq 0, \quad G_{ppp} \neq 0.$$

Note. This is the case $\lambda_x = 1$.

Theorem ([Mey70])

Let (x, s) be an extremal fixed point. Then,

- 1. There exists a family of fixed points x_t .*
- 2. x divides x_t into two arcs: one entirely consists of hyperbolic fixed points and the other consists of elliptic fixed points.*
- 3. In any coordinate system, s is a nondegenerate maximum or minimum of the family.*

Outline of Proof

We have an equation $G_p = G_q = 0$ at $(0, 0, 0)$, and the Jacobian is

$$\begin{pmatrix} 0 & 0 & G_{\eta t} \\ 0 & G_{\xi\xi} & G_{t\xi} \end{pmatrix}.$$

This has rank 2, so we have solution $(\xi, \eta, t) = (\xi(s), s, t(s))$ for the equation, i.e.

$$G_\xi(\xi(s), s, t(s)) = G_\eta(\xi(s), s, t(s)) = 0.$$

From this, we can deduce that $t'(0) = 0$ and $t''(0) \neq 0$, and

$$\begin{pmatrix} G_{\xi\xi} & G_{\eta\xi} \\ G_{\xi\eta} & G_{\eta\eta} \end{pmatrix} (\xi(s), s, t(s)) = \begin{pmatrix} G_{\xi\xi}(0) & 0 \\ 0 & sG_{\eta\eta\eta}(0) \end{pmatrix} + (\text{h.o.t})$$

Meyer's Theorem - Transitional Fixed Point

We call (x, s) is **transitional fixed point** if $\lambda_x = -1$ and satisfies some non-degeneracy condition. (See Appendix 2)

Theorem ([Mey70])

Let (x, s) be a transitional fixed point and x_t be an orbit cylinder in $U \times V \subset M \times S^1$. Then we can choose V small enough so that $V \setminus \{s\} = V_H \cup V_E$, where x_s is hyperbolic (elliptic) if $s \in V_H$ (V_E). Moreover, either one of the following holds.

- 1. There exists functions $x_1^H, x_2^H : V_H \rightarrow U$ such that $(x_i^H(t), t)$ is elliptic periodic points of period 2 and $x_i^H(t) \rightarrow x$ as $t \rightarrow s$,*
- 2. There exists functions $x_1^E, x_2^E : V_E \rightarrow U$ such that $(x_i^E(t), t)$ is hyperbolic periodic points of period 2 and $x_i(t) \rightarrow x$ as $t \rightarrow s$.*

We call these an arc of **nearby hyperbolic/elliptic periodic points**.

Meyer's Theorem - Bifurcation Points

Let $CM(x) = \{\exp(\pm 2\pi i l/k)\}$, $k \geq 3$, $(l, k) = 1$ and $0 < l < k$.

We call (x, s) is **k -bifurcation point** if it satisfies some non-degeneracy condition. (See Appendix 2.)

Theorem ([Mey70])

Let (x, s) be k -bifurcation point. Then for small orbit cylinder $U \times V$,

- 1. ($k = 3$) : There exists three arcs of nearby hyperbolic periodic points of period 3, parametrized by $V \setminus \{s\}$, converging to x .*
- 2. ($k \geq 5$) : There exists k arcs of nearby hyperbolic periodic points and k arcs of nearby elliptic periodic points of period k , parametrized by $V_+ \subset V \setminus \{s\}$, converging to x .*
- 3. ($k = 4$) : One of ($k = 3$) or ($k \geq 5$)-phenomena occurs.*

Let $\mathcal{G}$ be the set of 1-parameter family of symplectomorphisms on M such that

1. Each periodic point is elliptic, hyperbolic, extremal or transitional.
2. If $CM(x) = \{\exp(\pm 2\pi i l/k)\}$ for $k \geq 3$, then (x, s) is k -bifurcation point.

Theorem ([Mey70])

$\mathcal{G}$ is residual set in the space of 1-parameter family of symplectomorphisms.

$\Rightarrow$ Now we know the *generic* bifurcation behavior.

Classification of Generic Bifurcations

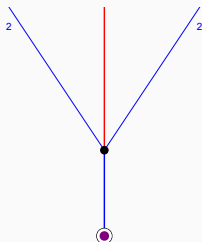
How to classify bifurcation

Let $\dim W = 4$, so the Poincaré section is 2-dimensional.

- For critical points, $CE(x) = \{\pm\alpha, \pm\bar{\alpha}\}$ in general.
- When there is $\alpha \in i\mathbb{R}$, Lyapunov theorem can be applied.
- For periodic orbits, if $\lambda_\gamma \neq 1$, there is an orbit cylinder.
- If $\lambda_\gamma^k = 1$, k -th cover of the orbit bifurcates.
- Using Poincaré section, a set of k periodic points of period k corresponds to a periodic orbit of period k .
- The nearby hyperbolic/elliptic arc on the Poincaré section corresponds to the orbit cylinder of hyperbolic/elliptic orbits.

Bifurcation Diagram

We'll see this kind of diagrams.



- Vertical axis is the energy.
- $\bullet$ is the bifurcation point.
- $\odot$ is the critical point of H .
- Each line is a family of periodic points of the return map in Poincaré section.
- Thick line is a family of fixed points, or 1-periodic orbits.
- Thin line is a family of k -periodic orbit, where the period is written aside the line. (2 here.)
- —, $\odot$ are elliptic, —, $\odot$ are hyperbolic, and $\odot$ is saddle.

List of Generic Bifurcations

No.	Name	Case	Tool
I	Burst / Reincarnation	x with $i\alpha \in CE(x)$	Lyapunov theorem
II	Creation / Annihilation	γ with $\lambda_\gamma = 1$	Extremal f.p.
III	Subtle Doubling / Halving	$\lambda_\gamma = -1$	Transitional f.p.
IV	Materialization / Murder	$\lambda_\gamma = -1$	Transitional f.p.
V	Phantom Kiss	$\lambda_\gamma^3 = 1$ $\lambda_\gamma^4 = 1$	3-bifurcation point 4-bifurcation point
VI	Emission / Absorption	$\lambda_\gamma^k = 1$ ($k \geq 4$)	k -bifurcation point
VII	Bubble / Liberation	Perturbation of critical point	TBA
VIII	Resonance	Perturbation of critical point	TBA

I. Burst / Reincarnation

Target. Critical point x with pure imaginary characteristic exponents.

Case 1. $CE(x) = \{i\alpha, r\}$ with $\alpha, r \in \mathbb{R}$. (saddle-center)



- By Lyapunov's theorem, we have a family of orbits near x
- The characteristic multipliers of the Lyapunov orbits near x is eventually real, so the orbits are hyperbolic.
- This hyperbolic family cannot be observed in the phase portrait, so we call this **phantom burst**

Figure 8: Phantom burst

I. Burst / Reincarnation

Case 2. $CE(x) = \{i\alpha, i\beta\}$ with $\mathbb{Q}$ -independent $\alpha, \beta \in \mathbb{R}$. (pure center)



Figure 9: Stable burst



Figure 10: Reincarnation

- By Lyapunov's theorem, we have two families near $\odot$.
- The characteristic multipliers are unimodular, so both orbits are elliptic.
- If two orbit cylinders are on the same side respect to the energy hyperplane, we observe two families of orbits burst from the critical point. (**Stable burst**)
- Otherwise, we observe a family of orbit absorbed into the critical point and burst out again. (**Reincarnation**)

II. Creation / Annihilation

Target. Periodic orbit such that $\lambda_\gamma = 1$.

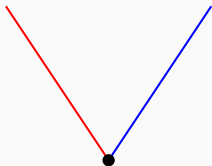


Figure 11: Creation and annihilation.
(Also called birth-death)

- The vector field develops an unstable periodic orbit $\bullet$. It's a closing of a recurrent orbit, called **Pugh catastrophe**.
- $\bullet$ splits into a hyperbolic orbit and an elliptic orbit.
- In the phase portrait, we observe an elliptic orbit suddenly appeared. (**Creation**)
- If we read this in reversed time, we can observe a vanishing pair of orbits. (**Annihilation**)

III. Subtle Doubling / Subtle Halving

Target. Periodic orbit such that $\lambda_\gamma = -1$.

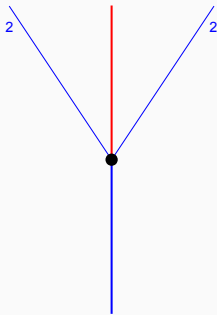
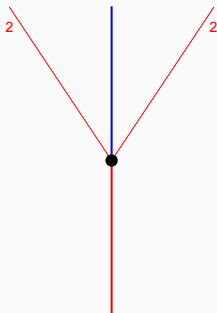


Figure 12: Subtle Doubling
(Also called period doubling)

- The elliptic orbit cylinder becomes hyperbolic at $\bullet$.
- An elliptic orbit with period 2 emerges.
- In phase portrait, we observe that the period of — is doubled. (**Subtle doubling**)
- In the opposite direction, the period is halved. (**Subtle halving**)

IV. Materialization / Murder

Target. Periodic orbit such that $\lambda_\gamma = -1$.

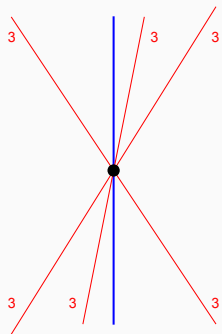


- The similar phenomena with subtle doubling, but the elliptic and hyperbolic are interchanged.
- In this case, we observe that — suddenly appeared. **(Materialization)**
- In the opposite direction, — suddenly disappears. **(Murder)**

Figure 13: Materialization and Murder

V. Phantom Kiss

Target. $\lambda_\gamma = \exp(2\pi il/k)$, $(l, k) = 1$ and $k = 3$ or 4 .

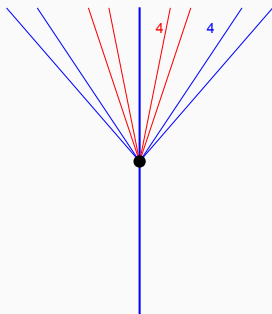


- A hyperbolic orbit of period 3 emerges in both directions of $\bullet$.
- We only observe *a moment of instability* of — .
- The family of hyperbolic orbit only experiences a sudden absorption to the elliptic orbit. **(Phantom kiss)**

Figure 14: Phantom Kiss

VI. Emission / Absorption

Target. $\lambda_\gamma = \exp(2\pi i l/k)$, $(l, k) = 1$ and $k \geq 4$.



- A hyperbolic and an elliptic orbit of period k both emerges at $\bullet$. (**Emission**)
- An elliptic orbit of period k is observed.
- In the opposite direction, k -periodic orbits are absorbed into the original periodic orbit. (**Absorption**)

Figure 15: Emission and Absorption

VII. Bubble / Liberation

Target. Fixed point 0 of parametrized Hamiltonian H_μ .

Assume that 0 is elliptic if $\mu < 0$ and hyperbolic if $\mu > 0$, i.e.

- $\mu < 0 : CE(x) = \{\pm i\alpha, \pm i\beta\}$ for $\alpha, \beta \in \mathbb{R}_{>0}$, $\alpha \neq \beta$.
- $\mu = 0 : \alpha = \beta$
- $\mu = 0 : CE(x) = \{z^{\pm 1}, \bar{z}^{\pm 1}\}$ where $z \notin S^1$.

If $\mu < 0$, the stable burst occurs, while nothing happens if $\mu > 0$.

Question. What happens to two elliptic orbit cylinders?

[MS71] showed that there are only two possibilities in general.

VII. Bubble / Liberation

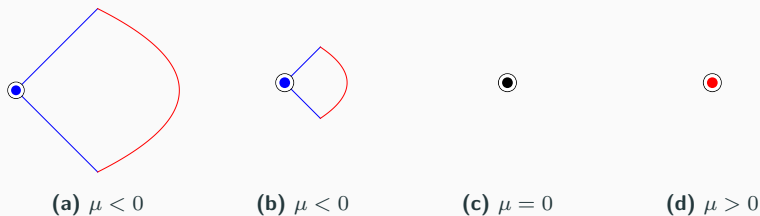


Figure 16: Bubble

The first possibility is a **bubble**: the shrink of two orbit cylinders into a fixed point.

VII. Bubble / Liberation

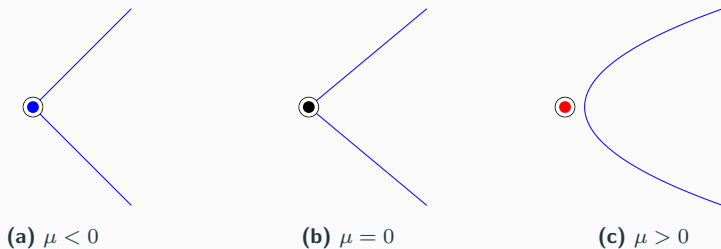


Figure 17: Liberation

The second possibility is a **liberation**: two orbit cylinders patch together. This phenomenon actually happens in CRTBP at L_4 .

VIII. Resonance

Target. Fixed point 0 of parametrized Hamiltonian H_μ .

For characteristic exponents α_1 and α_2 , if there exist coprime integers k, l such that

$$k\alpha_1 = l\alpha_2,$$

then $\mathcal{H}$ -elementary assumption of Lyapunov theorem is violated.

This is called **resonance**, and there are many possible different phenomenon occurs depending on (k, l) .

[SS73] gives the classification of generic resonance.

The situation is quite similar to the case of bubble: 'bridges' between two orbit cylinders are collapsed into a point.

A Life of Orbit

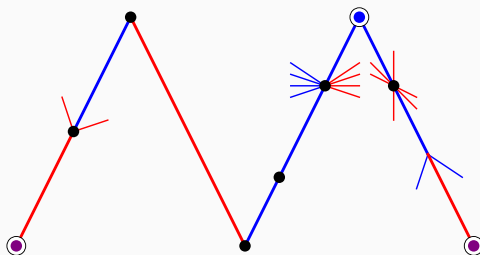


Figure 18: A life of an orbit

A periodic orbit can be born and die at critical point or other orbit.

For varying energy, it experiences various bifurcations, make birth of another orbits, and become from hyperbolic to elliptic or vice versa.

Closing

Genericity and Symmetry

Generically, a given Hamiltonian system is generic. (It's trivial.)

However, most of interesting systems are *not* generic, because of their symmetries in nature.

- The circular restricted three-body problem has natural $\mathbb{Z}_2$ -symmetry, a reflection along $q_1 q_2$ -plane.
- [Ayd23] The linear symmetries of Hill's lunar problem, a limit case of CRTBP, is given by $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$.
- The linear symmetries of the rotating Kepler problem is $S^1 \times S^1$.

Such symmetries prevent the bifurcation from being generic.

Many works were done for symmetric Hamiltonians, including [GSM87].

Pitchfork Bifurcation

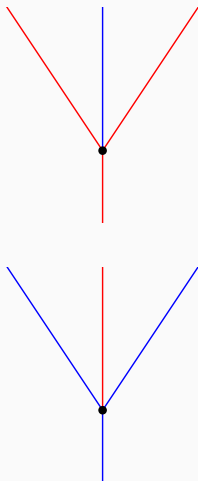


Figure 19: Pitchfork bifurcations

- The orbit changes hyperbolicity and ellipticity, and **two** orbits with **period 1** are born.
- The reason is $\mathbb{Z}_2$ -symmetry; if there is a new orbit on one side, then there must be another one on the other side.
- This type of bifurcation, which is not *generic*, is observed in Hill's lunar problem, CRTBP, and many other systems with symmetry.

Further Questions

Investigating bifurcation by Conley-Zehnder index:

- SH provides systematic way to study bifurcation.
- [AFvK⁺] studied Hill's lunar problem with CZ index.
- We might apply the method to three-body problem and other interesting systems to investigate periodic orbits and bifurcations.

General relation between the change of index and the bifurcation type:

- If we know the bifurcation type, what can we say about the index?
- If we detect index change, what can we say about bifurcation type?
- [FKM23] showed the invariance of χ_{SFT} explicitly in each case.

Thank you for your attention!

Appendix 1 : Fixed point and Generating Function

Let the 2nd-order part of G be

$$G(\xi, \eta) = \frac{1}{2}(\alpha\xi^2 + 2\beta\xi\eta + \gamma\eta^2),$$

and $4\Delta = \alpha\gamma - \beta^2 \neq -1$. The linear part of (Q, P) is given by

$$\begin{pmatrix} Q \\ P \end{pmatrix} = \frac{1}{1 + 4\Delta} \begin{pmatrix} (1 - \beta)^2 - \alpha\gamma & -2\gamma \\ 2\alpha & (1 + \beta)^2 - \alpha\gamma \end{pmatrix} \begin{pmatrix} q \\ p \end{pmatrix}.$$

The trace of given matrix is $T = 2(1 - 4\Delta)(1 + 4\Delta)^{-1}$.

- Min/Max of $G \Leftrightarrow \Delta > 0 \Leftrightarrow |T| < 2 \Leftrightarrow$ Elliptic fixed point
- Saddle of $G \Leftrightarrow \Delta < 0 \Leftrightarrow |T| > 2 \Leftrightarrow$ Hyperbolic fixed point

Appendix 2 - Non-degeneracy of bifurcation points

A fixed point (p, s) is **transitional** if locally

$$\begin{pmatrix} Q \\ P \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} q \\ p \end{pmatrix} + (\text{h.o.t}),$$

and for

$$P_{qq} = 2\alpha, \quad P_{qp} = \beta, \quad P_{qt} = \gamma, \quad P_{qqq} = 6\delta, \quad Q_{qq} = 2\zeta$$

with $\eta = \delta + \alpha\zeta + \alpha\beta/2$,

$$\gamma \neq 0, \quad \alpha^2 + 2\eta \neq 0.$$

Appendix 2 - Non-degeneracy of bifurcation points

Let $CM(p) = \{\exp(\pm 2\pi i l/k)\}$, $k \geq 3$, $(l, k) = 1$ and $0 < l < k$.

[Bir27] introduced a normalization procedure, which gives local expression

$$\begin{aligned}(q, p) &= (r^{1/2} \cos \theta, r^{1/2} \sin \theta), \\(Q, P) &= (R^{1/2} \cos \Theta, R^{1/2} \sin \Theta), \\ \Theta &= \theta + \frac{2\pi l}{k} + \frac{t\alpha(t)}{k} + \sum_{j=1}^{\lfloor (k-2)/2 \rfloor} \frac{\beta_j(t)}{k} r^j \\ &\quad + \frac{\gamma(t)}{k} r^{(k-2)/2} \cos k\theta + \Theta_1(\theta, r, t), \\ R &= r + \frac{2\gamma(t)}{k} r^{k/2} \sin k\theta + R_1(\theta, r, t).\end{aligned}$$

Appendix 2 - Non-degeneracy of bifurcation points

Moreover, we require for $r^{1/2} = \rho$,

$$\frac{\partial^j R_1}{\partial \rho^j}(\theta, 0, 0) = 0 \quad \text{for } j = 0, 1, \dots, k+1$$

$$\frac{\partial^i \Theta_1}{\partial \rho^i}(\theta, 0, 0) = 0 \quad \text{for } i = 0, 1, \dots, k-1.$$

We denote $\alpha(0) = \alpha$, $\beta_1(0) = \beta$, $\gamma(0) = \gamma$. If

- $(k = 3) \ \alpha, \gamma \neq 0$,
- $(k = 4) \ \alpha, \gamma, \beta \pm \gamma \neq 0$,
- $(k \geq 5) \ \alpha, \beta, \gamma \neq 0$,

we call (p, s) a **k -bifurcation point**.

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